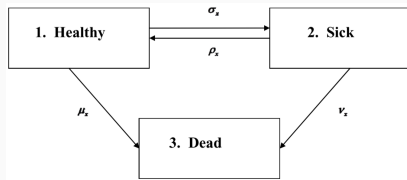


ACTL2102 - CTMC's - Inhomogenous CTMC's

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CTMC Revision



Remember in the CTMC model,

- q_{ij} represents the instantaneous probability we transition from i to j .
- The time we spend in a state is exponentially distributed with rate $\sum_{k \neq i} q_{ik}$.

Occupancy Probabilities

Recall we define $P_{ii}(t)$ the probability the process in state i is in state i after t time.

Definition (Occupancy Probabilities)

We define $P_{\bar{ii}}(t)$ the probability the process **remains** in state i over a period of t . This is equivalent to the time spent in the state exceeds t , so

$$P_{\bar{ii}}(t) = \mathbb{P}(T_i > t) = \exp \left\{ -t \sum_{k \neq i} q_{ik} \right\}.$$

The difference between $P_{ii}(t)$ is that we are allowed to transition out of i and come back, but in $P_{\bar{ii}}(t)$ we must remain there the entire time.

Motivating Inhomogenous CTMC's

So far, we have only considered stationary rates: the rate we transition from state i to state j at time t is q_{ij} (a constant).

But what if we want to have changing rates?

$\implies$ Just let q_{ij} be some function of t !

Generator Matrix

In doing this, our generator matrix becomes a function of t .

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Definition (Generator Matrix)

For a CTMC with rates q_{ij} , define

$$\mathbf{Q}(t) := \begin{pmatrix} q_{11}(t) & q_{12}(t) & \dots & q_{1n}(t) \\ q_{21}(t) & q_{22}(t) & \dots & q_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ q_{n1}(t) & q_{n2}(t) & \ddots & q_{nn}(t) \end{pmatrix}.$$

Note: Since $q_{ii}(t) = -\sum_{k \neq i} q_{ik}(t)$, the rows should sum to 0.

Definition

For a continuous time Markov chain X , define

$$P_{ij}(s, s + t) = \mathbb{P}(X(s + t) = j \mid X(s) = i).$$

Now, we need both arguments, as the process is no longer stationary.

Everything is still the same! If you can solve a question in the homogenous case, you have the skills to solve it in the inhomogenous case!

The whole 'find a path' approach is still valid, our constant rates are just replaced by functions.

Kolmogorov Equations

One small difference,

Theorem (Kolmogorov Equations)

For a homogenous Markov chain with state space S , for $t \geq 0$ we have

- Kolmogorov's Forward Equation

$$\frac{\partial}{\partial t} P_{ij}(s, t) = \sum_{k \in S} P_{ik}(s, t) q_{kj}(t),$$

- Kolmogorov's Backward Equation

$$\frac{\partial}{\partial s} P_{ij}(s, t) = - \sum_{k \in S} q_{ik}(s) P_{kj}(s, t).$$